

Data Analytics (PETR 6397)

Clustering

Dr. Ahmad Sakhaee-Pour

Petroleum Engineering Department

University of Houston

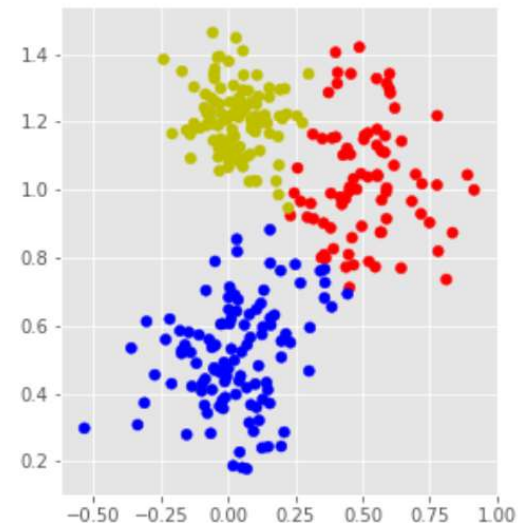
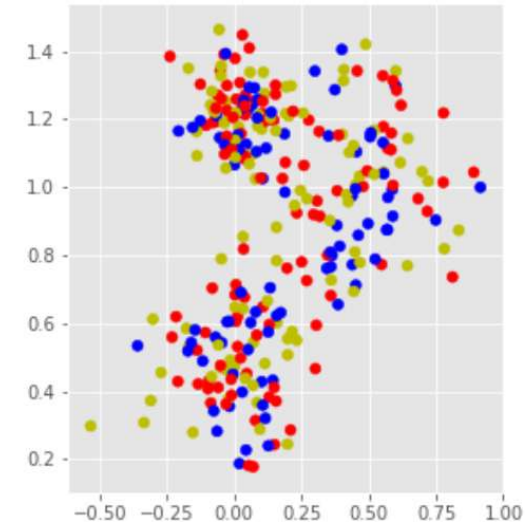
Spring 2025

Topics

- Unsupervised learning
- Unsupervised tasks
- Clustering:
 - *K*-mean clustering
 - Gaussian mixture model (GMM)
 - Hierarchical clustering
 - Density-based spatial clustering of applications with noise (DBSCAN)

Unsupervised learning

- Unsupervised model groups similar points together
- Similarity is based on distance, density, or other statistical measures
- Unsupervised learning is used for exploratory data analysis (EDA) to find main characteristics and draw inferences without labels



Unsupervised tasks (1)

- Clustering: Groups samples based on some assumptions (discussed later)
 - Example: customer segmentation, recommender system, search engine, and image segmentation
- Anomaly detection (outlier detection): The objective is to determine the “normal” data
 - Example: fraud detection, detecting defective products

Unsupervised tasks (2)

- Dimensionality Reduction: Reduces number of variables by finding principal variables
 - Example: Improving J function representation (bonus homework)
- Density estimation: Obtains the probability density function (PDF) of how most data are distributed
 - Example: heights of people based on their gender and nationality

Clustering vs. classification

- Similar to classification, clustering groups samples based on the similarity
- In contrast with classification, clustering does not have access to the correct labels

Non-exhaustive list of applications of clustering

- Customer segmentation: Groups customers based on their purchases and activities on the internet
- Feature engineering: The cluster affinity is useful as an extra feature(s). They improve the performance
- Image segmentation: Clusters pixels based on their colors (examples discussed in detail later)

Definition of cluster

- We can group data differently; there is no universal definition of a cluster

Common criteria

1. Gathering around a particular point (centroid)
2. Continuous regions of densely populated (cluster shape has more freedom than the previous one)
3. Hierarchical structures (clusters of clusters)

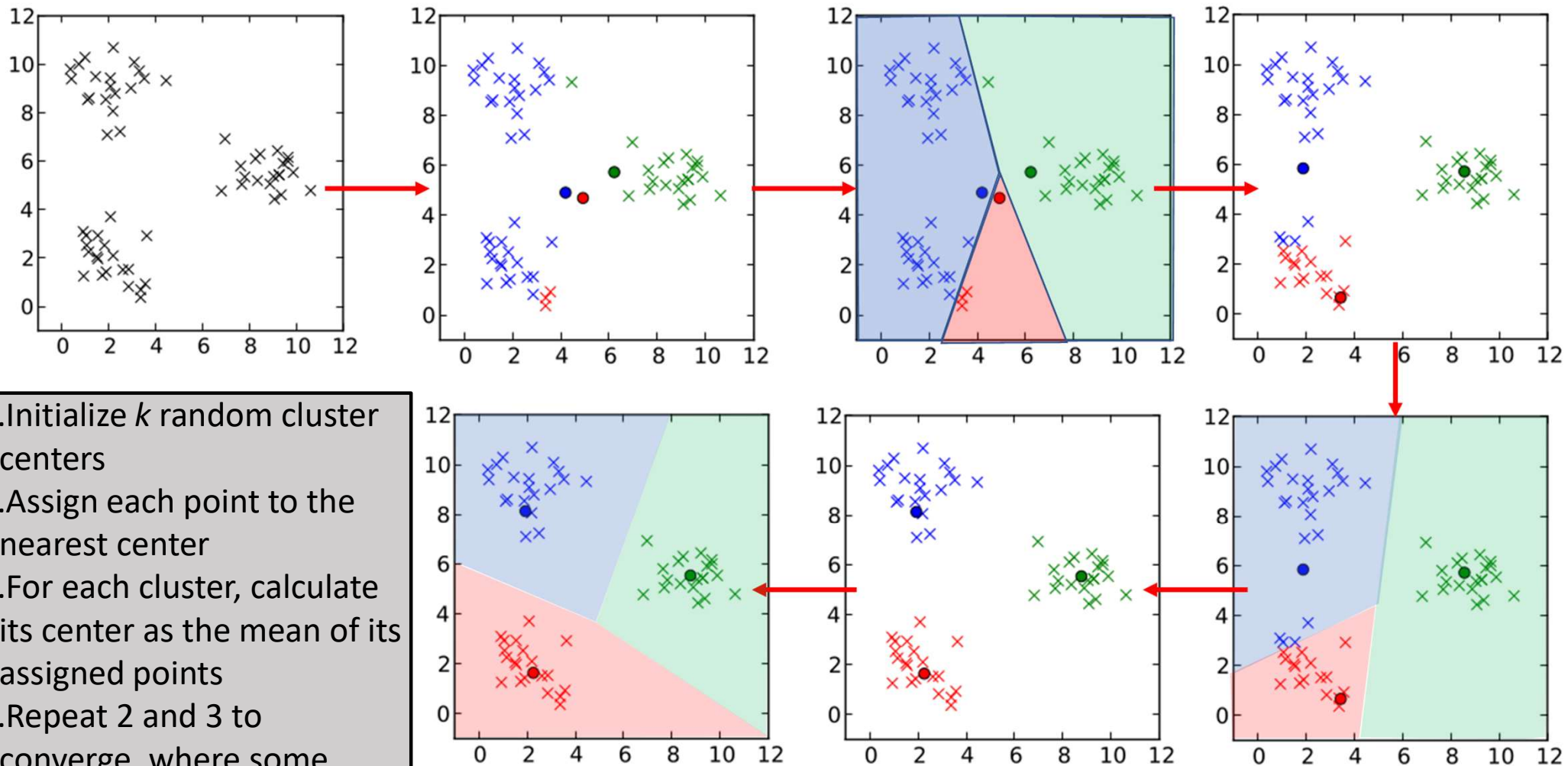
Clustering algorithms

K-means clustering

- This is the simplest and widely used clustering
- It divides data into different (disjoint) clusters, where each instance belongs to the cluster with the nearest mean

Pseudocode:

1. Initialize k random cluster centers
2. Assign each point to the nearest center
3. For each cluster, calculate its center as the mean of its assigned points
4. Repeat 2 and 3 to converge, where some criterion is satisfied



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Main characteristics of k -means clustering

- We have to define the number of clusters (k)
- K -means clustering converges quickly (it is one of the fastest clustering techniques)
- K -means clustering may not yield the best solution because we assigned the initial centroids randomly
- Potential solution: Run it a few times and choose the best outcome

Hard clustering vs. soft clustering

- Each instance has a label in k -means clustering (hard clustering)
- The label is NOT the class label (remember that clustering is not classification)
- Shortcoming: K -means clustering does not behave well when clusters have different diameters because it analyzes the distance of an instance from the centroid
- Potential solution: We may assign each instance a score per cluster (this is soft clustering)

Extensions of k -means clustering

- Regular k -means clustering initializes the clusters randomly and iterates using all instances to determine the clusters

Extended versions:

1. Smarter calculations of distances: Avoid unnecessary calculations of distances (2003) based on the triangle inequality
2. Different initialization: Use a smarter initialization, which is the basis of k -means++ (2006), rather than random
3. Mini-batch k -means: Use mini-batches of data (2010)

Evaluating the performance of clustering

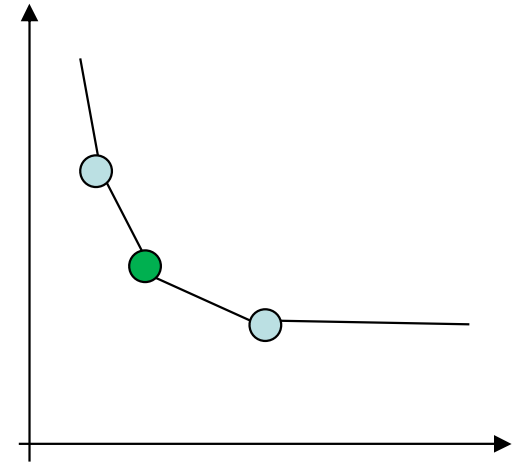
Elbow method for choosing the best number of clusters

1. Calculate Within-Cluster Sum of Squares (WCSS)

$$\text{WCSS} = \sum_i \sum_j \text{distance}(x_j, c_i)^2$$

x_j : instance j

c_i : centroid i



2. Find the cluster number where it is similar to an **elbow** (minus 45-degree tangent)

Q: What is WCSS when we assign every instance its own cluster?

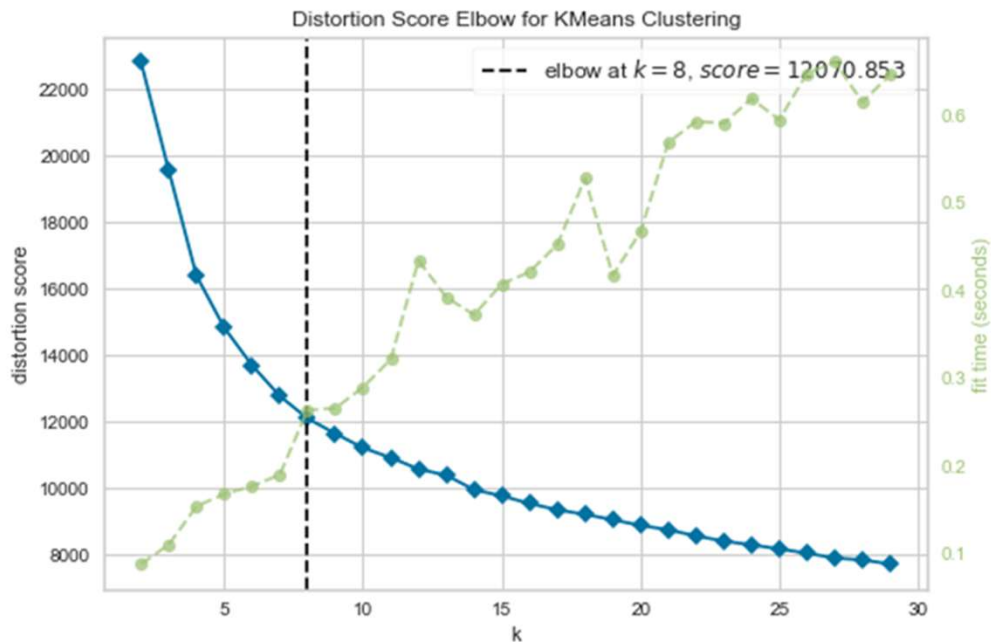
Silhouette coefficient

$$s(i)=[b(i)-a(i)]/\max(a(i),b(i))$$

- $a(i)$ is the distance between point i and other points in its cluster
- $b(j)$ is the smallest mean distance of point i to all points in any other cluster
- $-1 \leq s(i) \leq +1$
- Higher $s(i)$ means better clustering
- Silhouette score is the average of the Silhouette coefficient for all points (average of $s(i)$)

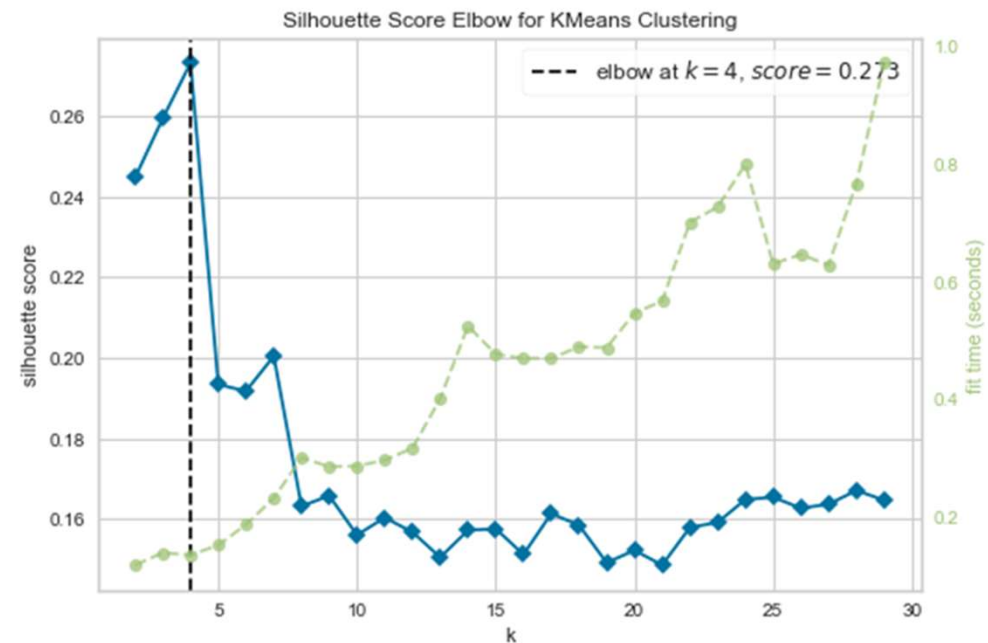
Select the optimal number of clusters

Elbow Method



8 Clusters

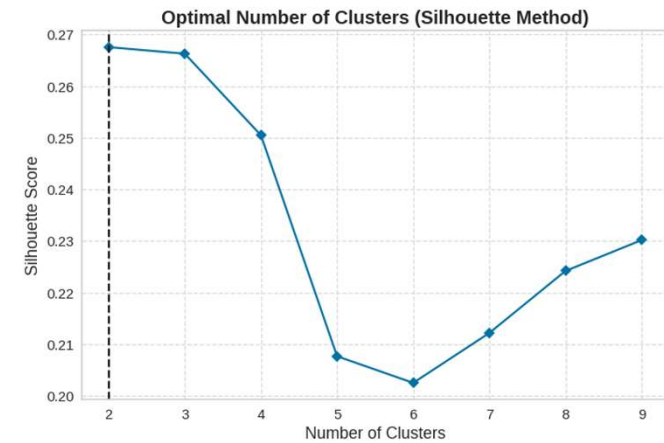
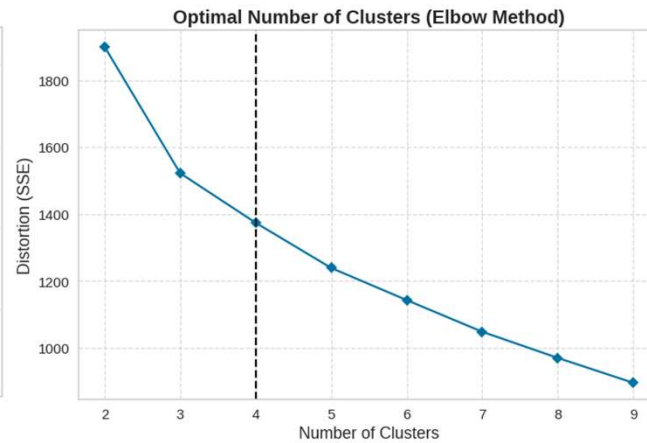
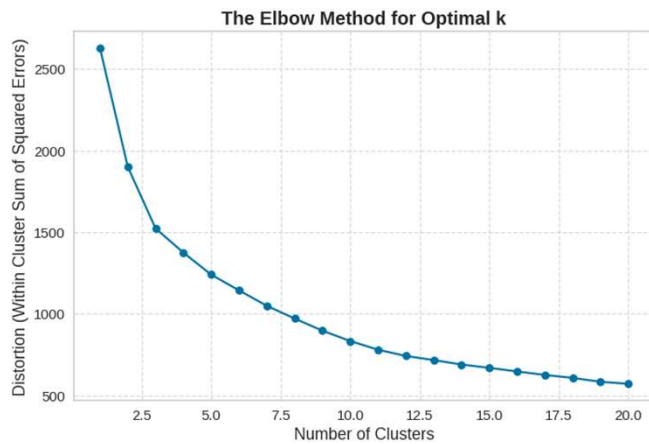
Silhouette Method



4 Clusters

Code: Let us try *k*-means clustering

- What is the optimal number of clusters? Why?

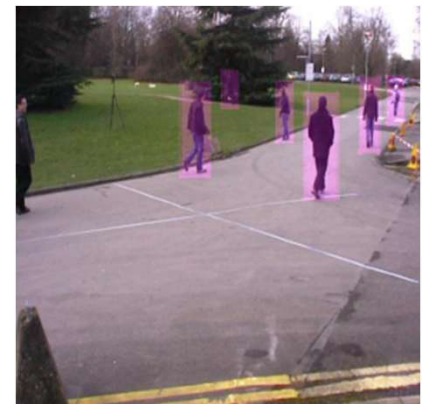


Limitations of K -means clustering

- k -means clustering performs poorly when clusters
 - have different sizes
 - have different densities
 - are non-spherical shapes
- Scaling improves the performance (but does not necessarily yield acceptable results)

Image segmentation is a common application of k -means clustering

- Color segmentation groups pixels with a similar color (example: determine which fraction of Canada's forest remained intact after wildfire)
- Semantic segmentation clusters pixels as part of an object (example: identify all pedestrians as a single group)
- Instance segmentation identifies the boundaries of individual objects (example: identify people separately)



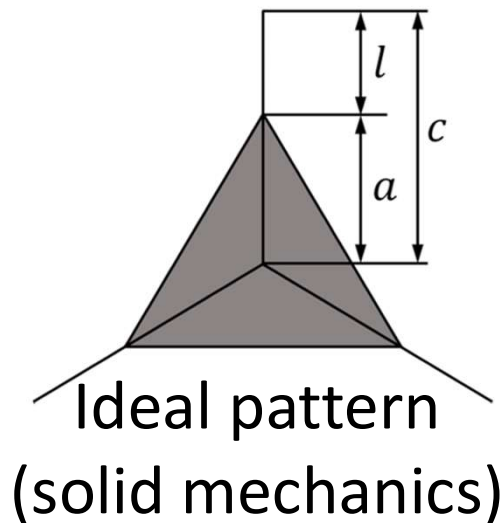
Ullah et al. 2018



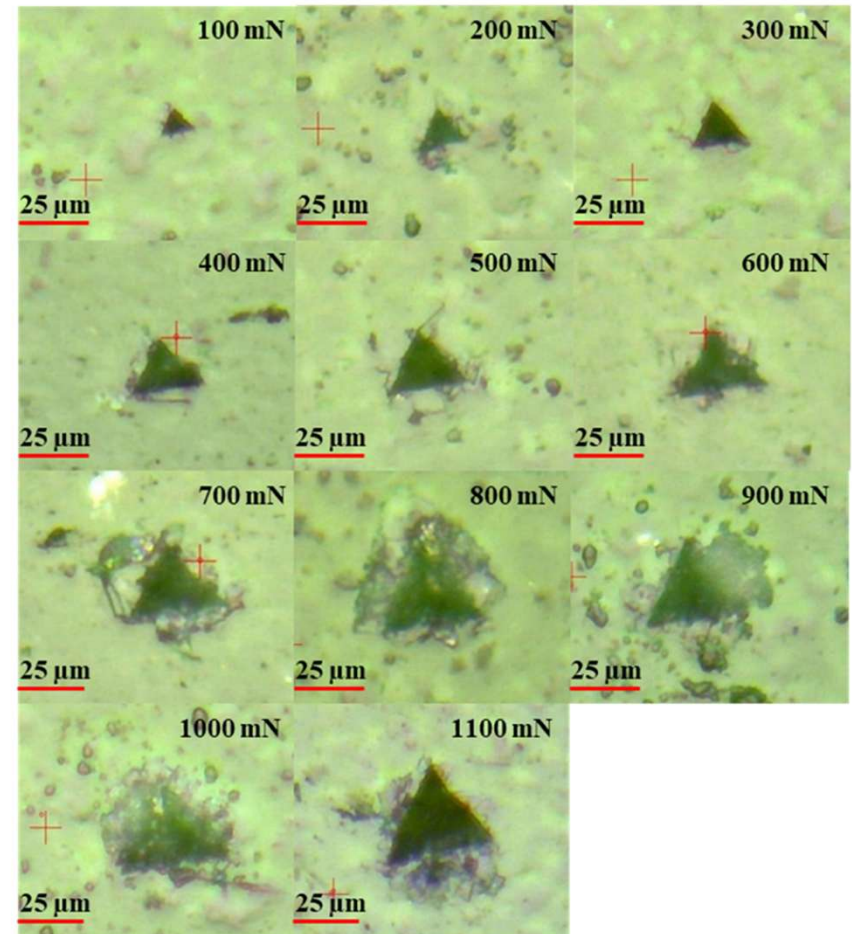
Bolya et al. 2019

Another example: Fracture toughness characterization of shale

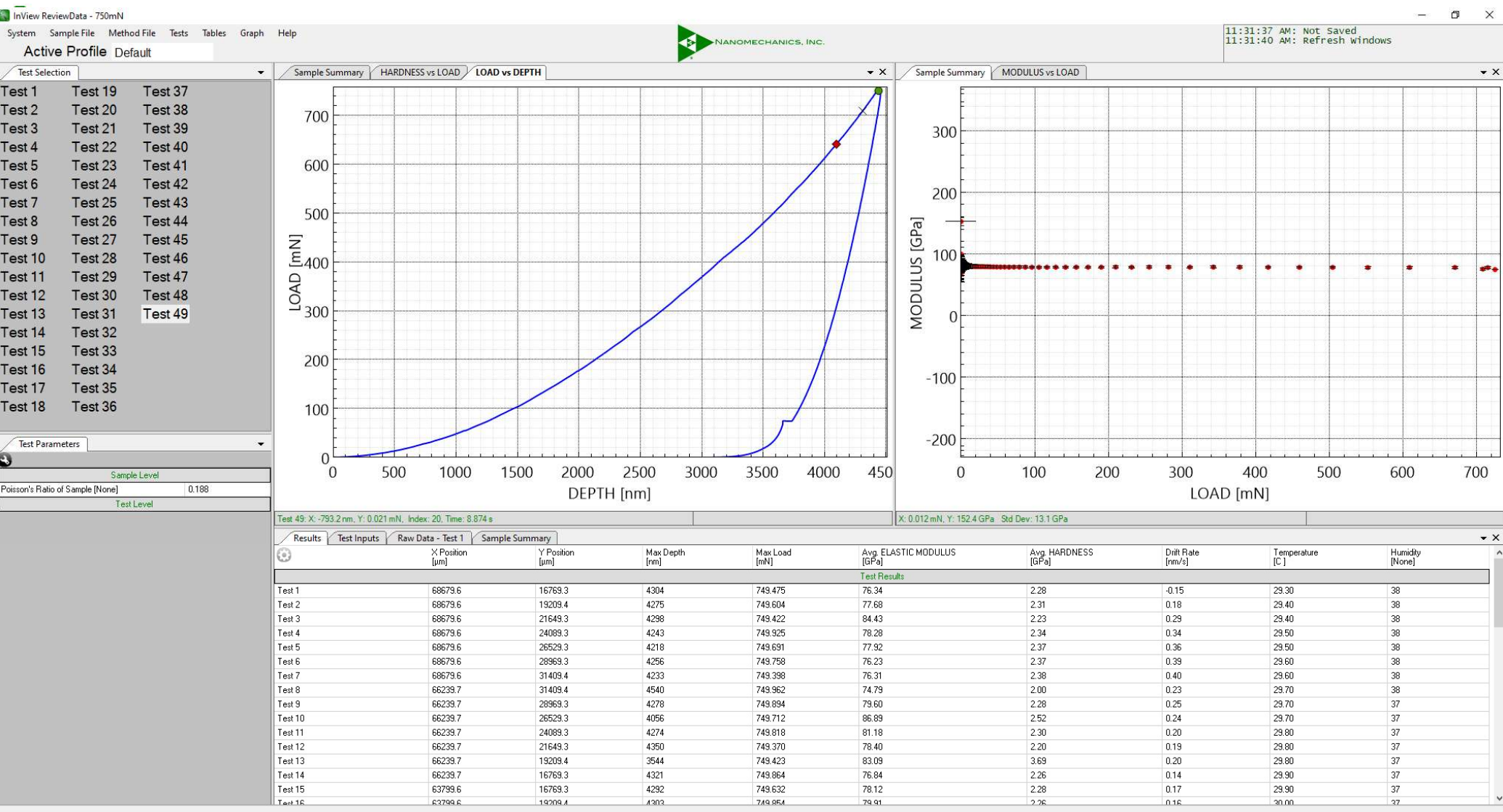
- Fracture toughness shows rock resistance against creating new fractures



$$K_c = \alpha \left(\frac{E_s}{H} \right)^{\frac{1}{2}} \frac{F_{\max}}{c^{\frac{3}{2}}}$$

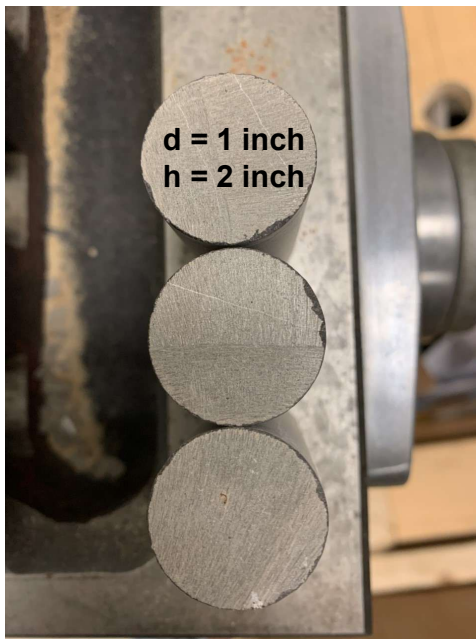


Actual pattern
(shale)



may

Sample preparation for standard triaxial test



Original sample

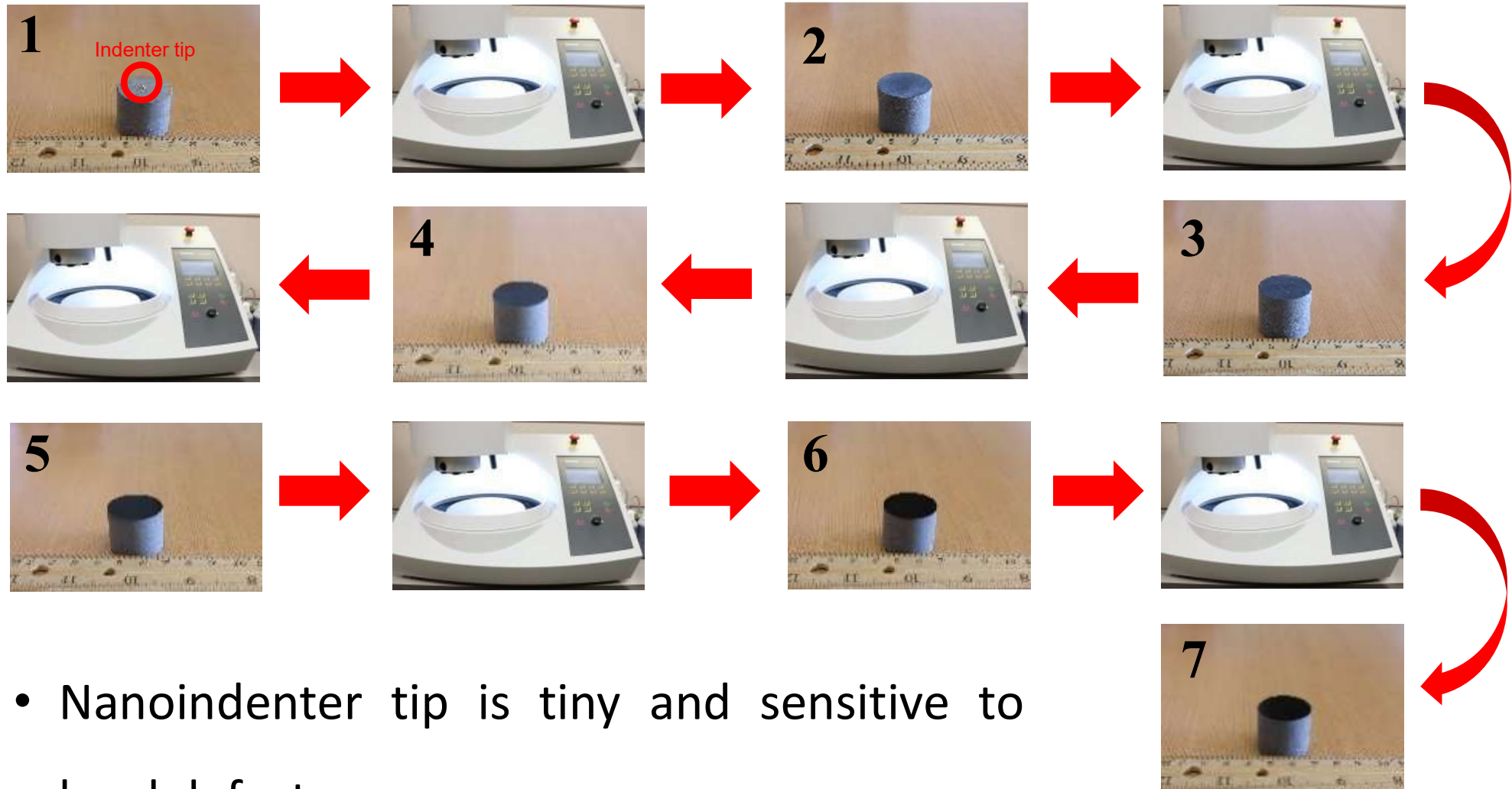


Preparation: ~ 10 minutes



Prepared sample

Sample preparation for nanoindentation



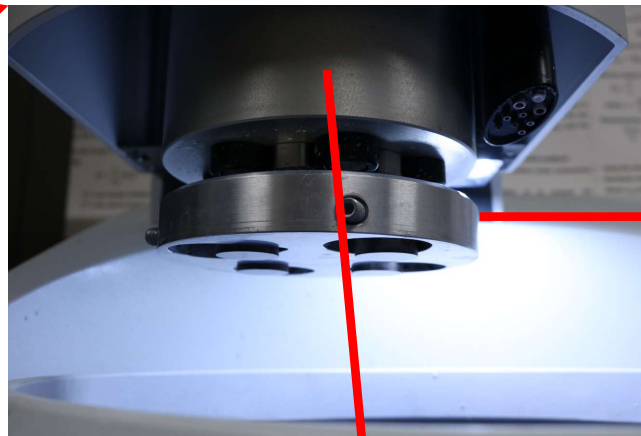
- Nanoindenter tip is tiny and sensitive to local defects

Surface polisher

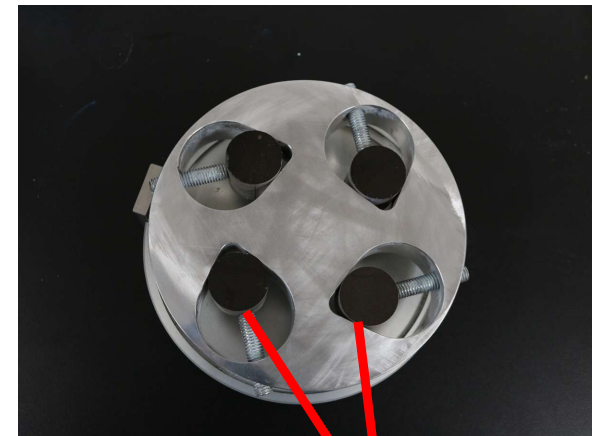


Mover plate

Suspensions

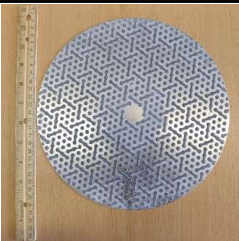
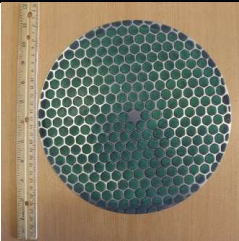
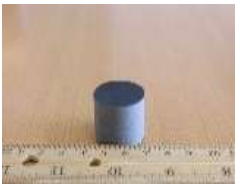
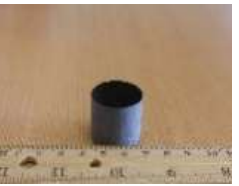
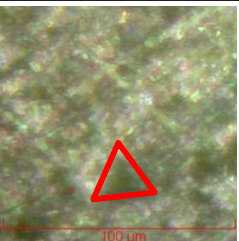
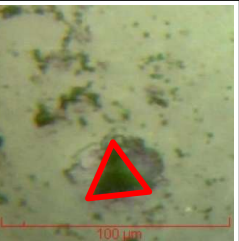


Mover head



Core plugs

Preparation discs for nanoindentation

Stages	Original (1)	MD-Piano 500 (2)	MD-Piano 1200 (3)	MD-Largo (4)	MD-Dac (5)	MD-Dur (6)	MD-Chem (7)
Disc Shape	-						
Time (mins)	0:00	1:00	1:00	5:00	4:00	4:00	2:00
Surface of Core Plug							
Results							

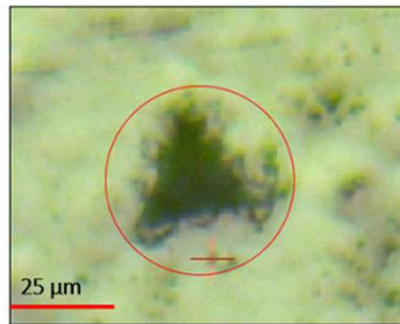
100 µm

Grinding

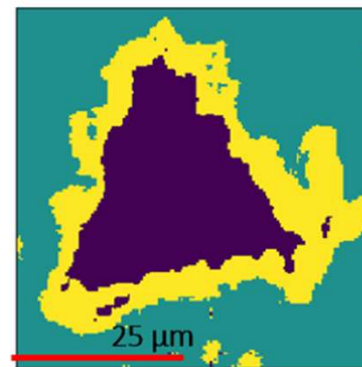
Polishing

Image segmentation allows us to analyze complex images of nanoindentation

- Each image has a bunch of pixels (example: MNIST gray image has 28×28 pixels)
- Color image has three channels: Red, Green, and Blue (RGB)



(a)



(b)

Alipour et al. 2021

To-do: read the reference to answer the listed questions

- What is active learning?
- What is uncertainty sampling?
- What is novelty detection?

Gaussian mixture model (GMM)

- It is based on the notion that data belong to a mixture of Gaussian distributions
- It is used for clustering
- It is also used for anomaly detection
- We must specify the number of Gaussian distributions (this is a generalization of k -means clustering)

GMM workflow

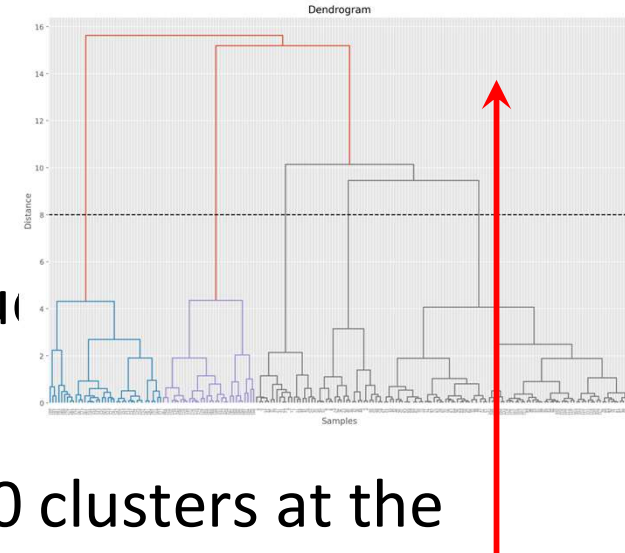
- It randomly initializes the parameters (mean and covariance) of the distributions
- Soft clustering: It assigns samples to the clusters randomly with some weights (this is different from *k*-means clustering)
- It uses expectation maximization (instead of distance minimization used in *k*-means clustering)

Hierarchical clustering

- Hierarchical clustering uses a distance-based algorithm to measure the distance between clusters (similar to k -means clustering)
- Two types of hierarchical clustering are discussed
 - Additive hierarchical clustering (agglomerative hierarchical clustering)
 - Divisive hierarchical clustering

Additive hierarchical clustering (agglomerative hierarchical clustering)

- This is a bottom-up approach
- First, every point will be assigned its unique cluster
- If there are 10 data points, there will be 10 clusters at the beginning
- Then, the closest pair of clusters are merged based on their distance
- This process is iterated to create a single cluster



Divisive hierarchical clustering (top-down)

- This is a top-down approach
- This is the opposite of additive hierarchical clustering
- First, all data points will be assigned to a single cluster
- The farthest point is then split in the cluster
- This process continues so each cluster has a single point

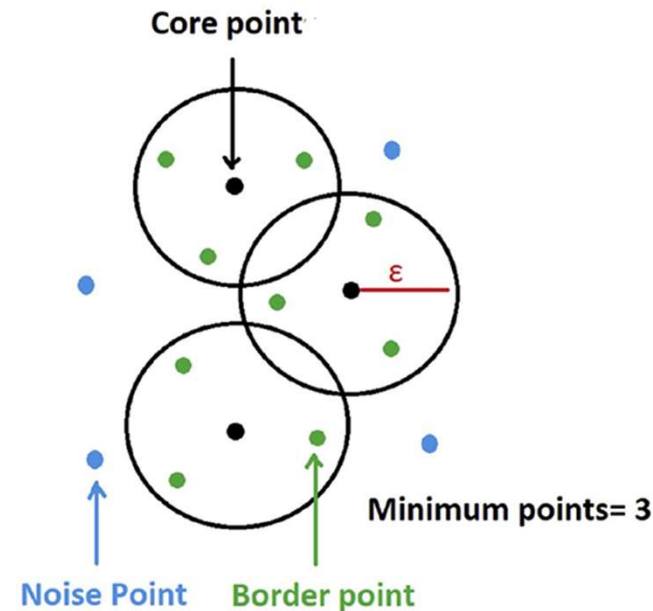


Density-based spatial clustering of applications with noise (DBSCAN)

- This approach divides the data into high-density and low-density regions
- It does not predict after training (we cannot use it to assign a new instance to a cluster)
- It has two tuning parameters (next slide)

Tuning parameters of DBSCAN

- ϵ : The radius of the circle that shows how close instances must be from one another to create a cluster
- MinPts: The minimum number of instances within the radius of a data point to be considered as a core point
- Core instance: An instance with at least MinPts in its ϵ neighborhood
- Noise (anomaly): An instance that is not a core instance and has no core instance in its ϵ neighborhood



Advantages of DBSCAN

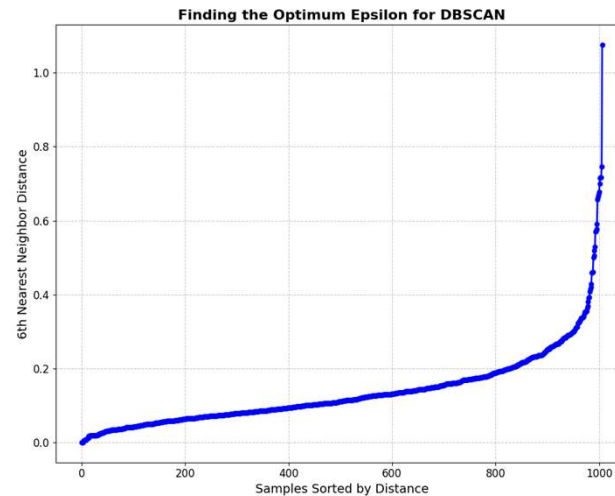
- It is robust to outliers (use this method instead of *k*-means clustering when you are unsure)
- It detects anomaly (noise)
- It can detect clusters with arbitrary shapes

Disadvantages of DBSCAN

- It performs poorly if clusters have very different densities
- It does not scale well to large datasets because of its computational cost

Code: Let us try DBSCAN

- Q1: What are the tuning parameters in DBSCAN?
- Q2: How does this code find epsilon?



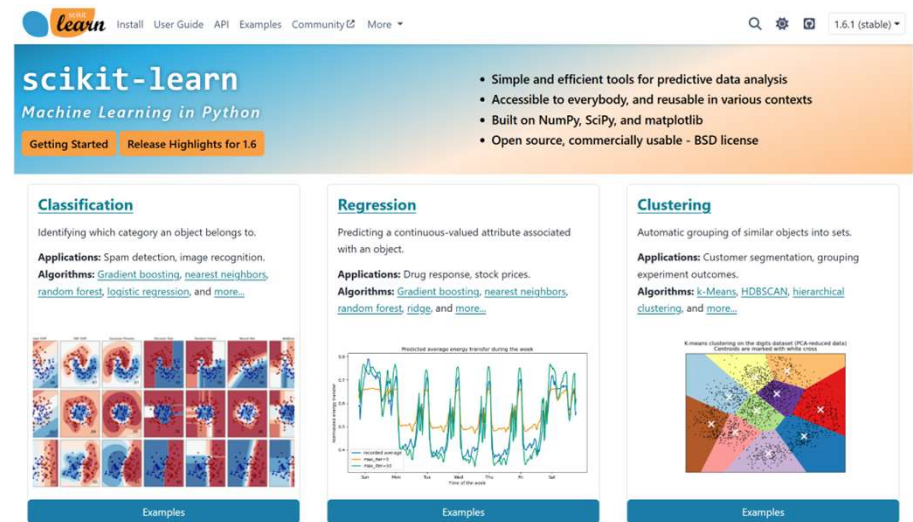
- Q3: How many clusters are created?
- Q4: How can we tune the parameters more realistically?

Two common clustering techniques

- *K*-means clustering is quite basic but works in most scenarios in practice
- DBSCAN helps you discard some samples as noise (this is useful in dealing with actual subsurface data with uncertainty)

Other clustering techniques

- BIRCH
- Mean-shift
- Affinity propagation
- Spectral clustering



Do not hesitate to try other clustering methods IF the common methods do not work for your data. You can compare the performance to pick the best model.